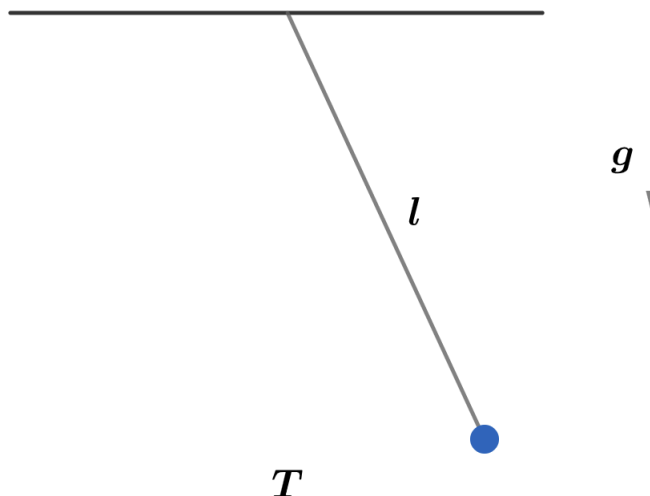


2023 F=ma Exam: Problem 19

Kevin S. Huang



The period of a pendulum is given by

$$T = 2\pi\sqrt{\frac{l}{g}}$$

so we have

$$g = \frac{4\pi^2 l}{T^2}$$

Recall relative uncertainties add in quadrature when independent variables are multiplied:

$$\frac{\Delta g}{g} = \sqrt{\left(\frac{\Delta l}{l}\right)^2 + \left(\frac{\Delta T^{-2}}{T^{-2}}\right)^2}$$

Recall the relative uncertainty is multiplied by the exponent when a variable is raised to a power:

$$\frac{\Delta T^{-2}}{T^{-2}} = |-2| \frac{\Delta T}{T}$$

Thus,

$$\Delta g = g \sqrt{\left(\frac{\Delta l}{l}\right)^2 + \left(\frac{2\Delta T}{T}\right)^2} = (10.1 \text{ m/s}^2) \sqrt{\left(\frac{0.01 \text{ m}}{0.50 \text{ m}}\right)^2 + \left(\frac{2(0.1 \text{ s})}{1.4 \text{ s}}\right)^2} = 1.4 \text{ m/s}^2$$

so the answer is C.